



GENERATIVE AI MASTERCLASS

From Beginner to Advanced – Build, Automate & Innovate with Generative AI

Master the complete Generative AI ecosystem—from AI fundamentals and prompt engineering to LLMs, AI agents, RAG, multimodal AI, and enterprise AI application development. Gain practical experience by building real-world AI solutions using today's leading tools and frameworks.



Course Roadmap (Beginner → Advanced)

01. AI & Generative AI Fundamentals

- Introduction to Artificial Intelligence
 - Machine Learning vs Deep Learning
 - Generative AI Concepts
 - Large Language Models (LLMs)
 - AI Terminology
 - AI Applications Across Industries
 - AI Ethics & Responsible AI
 - Future of AI
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02. Prompt Engineering

- Prompt Design Principles
 - Zero-Shot Prompting
 - One-Shot & Few-Shot Prompting
 - Chain-of-Thought Prompting
 - Role-Based Prompting
 - Prompt Templates
 - Prompt Optimization
 - AI Productivity Techniques
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03. Large Language Models (LLMs)

- LLM Architecture
- Tokens & Embeddings
- Context Windows

- Fine-Tuning Concepts
 - Model Selection
 - Open-Source vs Proprietary Models
 - Model Evaluation
 - Cost & Performance Optimization
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04. AI Tools & Productivity

- ChatGPT
 - Microsoft Copilot
 - Google Gemini
 - Claude AI
 - Perplexity AI
 - NotebookLM
 - AI for Research
 - AI for Content Creation
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05. AI Development with Python

- Python Basics for AI
 - OpenAI SDK
 - Hugging Face Transformers
 - API Integration
 - AI Automation Scripts
 - Environment Setup
 - Jupyter Notebook
 - Building AI Utilities
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06. Retrieval-Augmented Generation (RAG)

- RAG Architecture
- Document Loading
- Text Chunking
- Embeddings
- Vector Databases
- Semantic Search
- Retrieval Pipelines
- Enterprise Knowledge Assistants

07. AI Agents & Workflow Automation

- AI Agent Fundamentals
 - Agentic AI Concepts
 - Multi-Agent Systems
 - Tool Calling
 - Function Calling
 - Workflow Automation
 - AI Assistants
 - Business Process Automation
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08. Multimodal AI

- Image Generation
 - Image Understanding
 - Speech-to-Text
 - Text-to-Speech
 - Video AI
 - OCR & Document AI
 - Vision-Language Models
 - Multimodal Applications
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09. Build & Deploy AI Applications

- LangChain
 - LlamaIndex
 - FastAPI
 - Streamlit
 - Gradio
 - AI Chatbots
 - AI API Deployment
 - Cloud Deployment Basics
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Hands-on Projects

- AI Chatbot
- PDF Q&A Assistant
- Resume Analyzer

- AI Content Generator
 - AI Research Assistant
 - Document Summarizer
 - Customer Support Bot
 - AI Email Assistant
 - RAG Knowledge Base
 - AI Agent Workflow Automation
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Tools & Technologies

- ChatGPT
 - Microsoft Copilot
 - Google Gemini
 - Claude AI
 - Perplexity AI
 - NotebookLM
 - Python
 - OpenAI API
 - Hugging Face
 - LangChain
 - LlamaIndex
 - FastAPI
 - Streamlit
 - Gradio
 - ChromaDB / FAISS
 - Git & GitHub
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Career Opportunities

- Generative AI Engineer
- AI Application Developer
- Prompt Engineer
- AI Solutions Engineer
- AI Consultant
- LLM Application Developer
- AI Automation Engineer
- AI Product Specialist
- AI Innovation Lead

- AI Solutions Architect
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★ Course Highlights

-  Beginner to Advanced Curriculum
-  Hands-on AI Projects
-  Prompt Engineering Mastery
-  LLM & RAG Development
-  AI Agents & Workflow Automation
-  Multimodal AI Applications
-  Python for AI Development
-  Portfolio-Ready Projects
-  Industry Best Practices
-  Interview & Career Preparation

Tagline

"Master Generative AI. Build Intelligent Applications. Shape the Future with AI."